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Skill 21.1 Exercise 1

The two code segments below both open, read, and close the file *document.txt*.

Segment 1	Segment 2
<pre>my_doc = open('document.txt') text = my_doc.read() my_doc.close() print(text)</pre>	<pre>with open('document.txt') as my_doc: text = my_doc.read() print(text)</pre>

What are the advantages of *Segment 2* over *Segment 1*?

Skill 21.2 Exercise 1

Refer the text which is stored in *some_lines.txt*

some_lines.txt

1. How many lines do we write on the daily
2. Many money, we write many many many
3. How many lines do you write on the daily
4. Say you say many money, you write many many many

Use a *with* statement to create a file object pointing to the file *some_lines.txt* shown to the right. Store that file object in the variable *lines_doc*.

Assume what you write above is correct, what is printed for each of the following statements.

```
print(lines_doc)
print(lines_doc.readlines())
```

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Iterate through each of the lines in `lines_doc.readlines()` using a for loop.
Inside the for loop print out each line of *some_lines.txt*.

Skill 21.2 Exercise 2

Refer the text which is stored in *some_lines.txt*

some_lines.txt

1. How many lines do we write on the daily
2. Many money, we write many many many
3. How many lines do you write on the daily
4. Say you say many money, you write many many many

Use a *with* statement to create a file object pointing to the file *some_lines.txt* shown to the right. Store that file object in the variable *lines_doc*.

Assume what you write above is correct, what is printed for each of the following statements.

```
lines_doc.readline()
lines_doc.readline()
lines_doc.readline()
print(lines_doc.readline())
```

Skill 21.3 Exercise 1

Create a file object for the file *bad_bands.txt* using the *open()* function with the *w* argument. Assign this object to the temporary variable *bad_bands_doc*.

Use the *bad_bands_doc.write()* method to add the name of a musical group you dislike to the document *bad_bands.txt*.

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Skill 21.4 Exercise 1

In a previous exercise, we created a file called *bad_bands.txt*.

Write two additional bad bands to this file by using the append argument, 'a'. Each band should be added on a separate line.

Skill 21.5 Exercise 1

Refer to the csv file below,

users.csv

```
Name,Username,Email
Roger Smith,rsmith,wigginsryan@yahoo.com
Michelle Beck,mlbeck,hcosta@hotmail.com
Ashley Barker,a_bark_x,a_bark_x@turner.com
Lynn Gonzales,goodmanjames,lynniegonz@hotmail.com
Jennifer Chase,chasej,jchase@ramirez.com
Charles Hoover,choover,choover89@yahoo.com
Adrian Evans,adevans,adevans98@yahoo.com
Susan Walter,susan82,swilliams@yahoo.com
Stephanie King,stephanieking,sking@morris-tyler.com
Erika Miller,jessica32,ejmillier79@yahoo.com
```

In the space below do the following,

- Import the csv module
- Create an empty list called *user_emails*
- Open up the file *users.csv* in the temporary variable *users_file*.
- Use the *csv.DictReader* method to read the contents of *users_file* into a new variable called *users_dict*.
- Use a loop to iterate over each *users_dict* and store the email for each user in *user_emails*

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Skill 21.6 Exercise 1

Refer to the csv file below,

books.csv

Author@ISBN@Title

Lauren Murray@978-0-12-995015-8@"Enviornment Call, Amount Later Page Country"

Micheal Jones@978-1-78110-100-1@Rate Security Full

Alexander Carr@978-0-315-25137-3@Still Response Size

Michael Williams@978-0-388-70665-7@Position Result Five

Kathleen Ferguson@978-1-75098-721-6@Country Week Receive And Sign

Sarah Dorsey@978-1-06-483628-6@Audience Truth Small

Mary Middleton@978-0-7419-8114-1@Travel: Special Offer Near Allow Goal

William Todd@978-1-4457-0480-7@Money Exactly Drop Teach

Joan Martin@978-0-657-61030-2@Theory Do Half Change

Gary Roman@978-1-5039-7539-2@Bill Serve Pull Industry South Job

In the space below do the following,

- Import the csv module
- Open up the file *books.csv* in the temporary variable *books_file*.
- Use the *csv.DictReader* method to read the contents of *books_file* into a new variable called *books_dict* (what delimiter is being used here?).
- Create a list called *isbn_list*
- Use a loop to iterate over each *books_dict* and store the ISBN for each book in *isbn_list*

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Skill 21.7 Exercise 1

Refer to the list of access log dictionaries below,

```
access_log = [{ 'time': '08:39:37', 'limit': 844404, 'address': '1.227.124.181'},  
{ 'time': '13:13:35', 'limit': 543871, 'address': '198.51.139.193'}, { 'time':  
'19:40:45', 'limit': 3021, 'address': '172.1.254.208'}, { 'time': '18:57:16',  
'limit': 67031769, 'address': '172.58.247.219'}, { 'time': '21:17:13', 'limit':  
9083, 'address': '124.144.20.113'}, { 'time': '23:34:17', 'limit': 65913, 'address':  
'203.236.149.220'}, { 'time': '13:58:05', 'limit': 1541474, 'address':  
'192.52.206.76'}, { 'time': '10:52:00', 'limit': 11465607, 'address':  
'104.47.149.93'}, { 'time': '14:56:12', 'limit': 109, 'address': '192.31.185.7'},  
{ 'time': '18:56:35', 'limit': 6207, 'address': '2.228.164.197'}]
```

```
fields = ['time', 'address', 'limit']
```

In the space below, do the following

- Import the csv module
- Open up the file *logger.csv* in the temporary variable *logger_csv*. Don't forget to open the file in write-mode.
- Create a *csv.DictWriter* instance called *log_writer*. Pass *logger_csv* as the first argument and then *fields* as a keyword argument to the keyword *fieldnames*.
- Write the header to *log_writer* using the *.writeheader()* method.
- Iterate through the *access_log* list and add each element to the CSV using *log_writer.writerow()*.

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Skill 21.8 Exercise 1

Refer to the json file below,

message.json

```
{'text': "Now that's JSON!", 'secret text': "Now that's some _serious_ JSON!"}
```

In the space below, do the following,

- Import the json module
- Open up the file *message.json* and save the file object to the variable *message_json*.
- Pass the JSON file object as an argument to *json.load()* and save the resulting Python dictionary as *message*.
- Print out *message['text']*.

What is the output from the print statement above?

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Skill 21.9 Exercise 1

Refer to the dictionary below,

```
data_payload = {'interesting message': 'What is JSON? A web application\'s little  
pile of secrets.',  
                'follow up': 'But enough talk!'}
```

In the space below do the following,

- Import the json module
- Open a new file object in the variable *data_json*. The filename should be *data.json* and the file should be opened in write-mode.
- Call *json.dump()* with *data_payload* and *data_json* to convert our data to JSON and then save it to the file *data.json*.